Reading Environment and History through Tree Rings

Lesson 4 Sampling, measurement and ring-width series · 90 minutes

Plan sampling, preserve specimens and records, observe and measure boundaries, verify dating, then detrend and build a chronology. Return to the wood and records when a mismatch appears.

Sampling and records

Choose species, sites, trees and spatial coverage for the question. Tree and radius counts depend on the material and purpose.

Link site, tree, specimen and path IDs to collection date, position, species, pith, bark, losses, photographs and processing conditions.

Preserve orientation and fragment order. Material such as waterlogged wood needs suitable conservation.

Mini exercise: moisture variability in a forest with ridges and valleys

Sites, trees and radii to sample, with reasons:

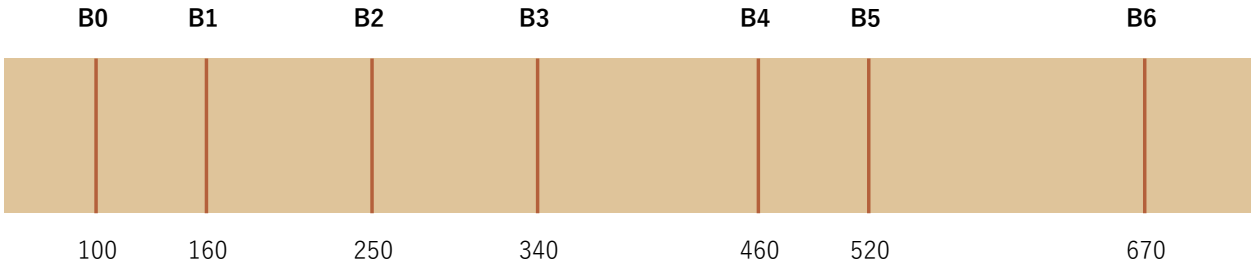
Metadata to retain:

A limitation arising from the areas not sampled:

All numerical data and diagrams in this lesson are synthetic teaching examples, not empirical observations.

Exercise A From boundary positions to widths

A known 10 mm spans 600 px. Boundary positions are 100, 160, 250, 340, 460, 520 and 670 px, increasing from inner to outer wood.



Boundary positions (px) · schematic, not a wood image

Spatial scale: 600 ÷ 10 = _____ px/mm

Ring	Boundaries	Difference (px)	Width (mm)
1	B0–B1		
2	B1–B2		
3	B2–B3		
4	B3–B4		
5	B4–B5		
6	B5–B6		

Width (mm) = successive boundary separation (px) ÷ spatial scale (px/mm).

Sum of six widths: _____ mm / Endpoint distance: (670 − 100) ÷ scale = _____ mm

Why can a matching total fail to prove correct boundary identification?

Check the scale in the specimen plane and account for resizing or distortion. Pixel count and displayed decimals alone do not establish accuracy.

Quality control and dating records

Check units, precision, orientation, duplicate or missing rows, extremes and jumps. Investigate flagged values in the image and specimen before revising them.

Specimen / path / image file:

Ring or boundary position to inspect:

Observed evidence:

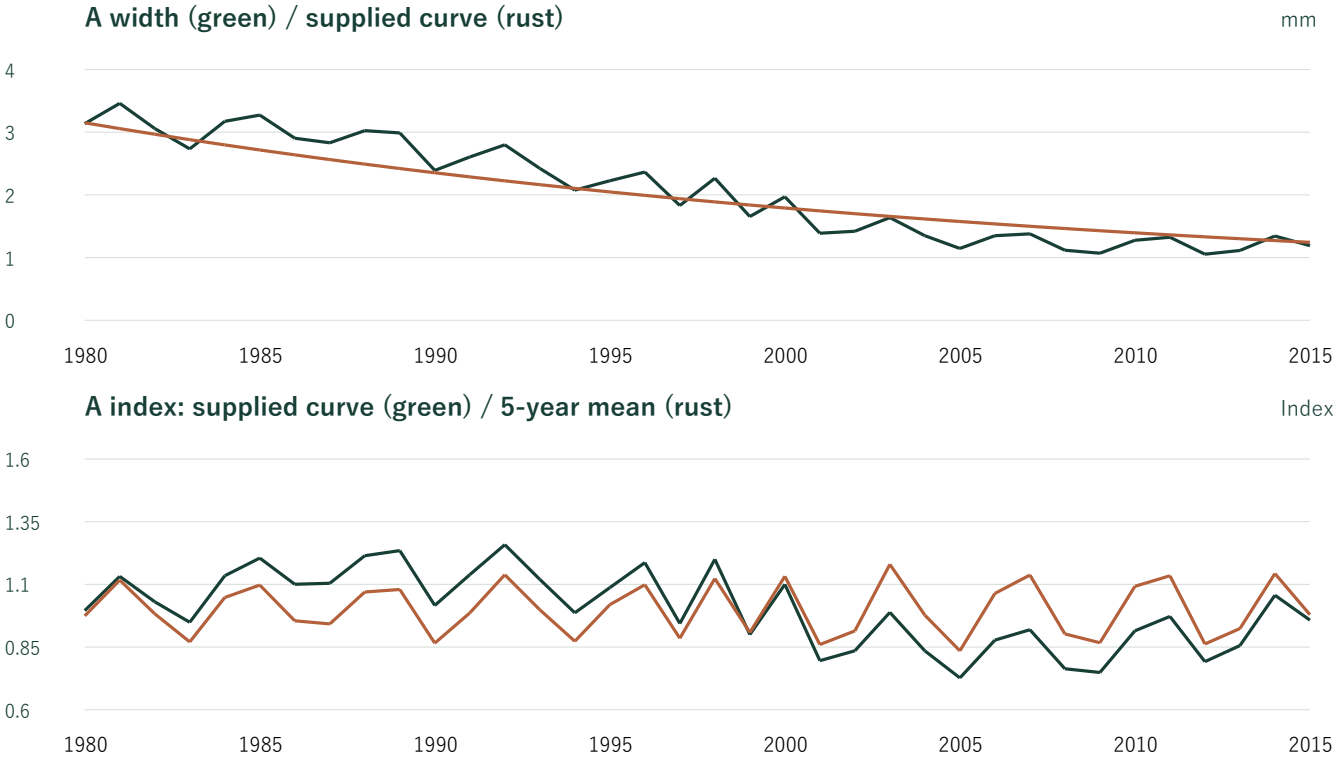
Possible causes and additional evidence needed:

Any revision, its reason, operator and date:

Distinguish ring numbers from calendar years. Investigate local absences, false rings and missed boundaries using anatomy and independent comparisons. Preserve observations and record dating hypotheses separately.

What does standardization retain?

In this exercise, a ring-width index is width divided by a supplied growth-curve value. Both inputs simulate mm; the ratio is dimensionless. Real analyses choose detrending methods for their question.



The comparison uses a centred five-year mean, with available years at the edges. It demonstrates sensitivity rather than a recommended setting; a flexible reference can remove longer-term variation.

How could the information you need to retain depend on the research question?

Exercise B Build indices and a chronology

Open L04_B_widths_and_supplied_curves.csv. A contains years, B/C the width/curve of A, D/E those of B, and F/G those of C. Add indices in H:J, their available-value mean in K and tree count in L.

H2 = IF(ISNUMBER(B2), B2/C2, "") ／ I2 = IF(ISNUMBER(D2), D2/E2, "") ／ J2 = IF(ISNUMBER(F2), F2/G2, "")

K2 = AVERAGE(H2:J2) ／ L2 = COUNT(H2:J2)

Fill down from row 2 through row 37. Preserve blanks. The excerpt below also permits calculation without a computer. Divide width by the supplied curve.

Year	Tree	Width mm	Curve mm	Index
1998	A	2.26560	1.88800	
1998	B	1.92698	1.60582	
1998	C	1.09175	1.21306	
1999	A	1.65469	1.83854	
1999	B	1.24045	1.55056	
1999	C	1.18460	1.18460	
2000	A	1.96986	1.79078	
2000	B	1.94705	1.49773	
2000	C	—	—	

Year	Mean index	Trees n
1998		
1999		
2000		

What changes if C in 2000 becomes zero? What further checks would you want where fewer trees are available?

Terms and linked resources

Japanese / English	Brief explanation
成長錐試料（increment core）	A cylindrical radial wood sample.
メタデータ（metadata）	Information describing specimens and processing.
測線（measurement path）	The path used to measure widths.
空間較正（spatial calibration）	Converting pixel distance to physical length.
品質管理（quality control (QC)）	Checking observations and numerical consistency.
標準化（standardization）	Adjusting growth levels or trends for the question.
傾向除去（detrending）	Expressing variation relative to a chosen trend.
年輪幅指数（ring-width index）	Here, the ratio of width to the supplied curve.
年表（chronology）	A series combined by calendar year.
試料数（sample depth）	Available samples by year; distinguish tree counts.

R26 Sample preparation and measurement

R29 Spatial calibration in ImageJ

R27 Tree-ring data and processing

R28 dplR: detrending and chronologies

R08 International Tree-Ring Data Bank

R09 Glossary of Dendrochronology

Foundational sources: Speer (2010), Chapter 5, and the prior Turku field, lab and standardization lectures. Figures and numerical data are newly composed. No source figures are reproduced.

Lesson 4 Calculate ring-width indices on paper

For Exercise B on slide 24. About six minutes; no computer or CSV import is needed. These are synthetic values for practising the calculation, separate from the measured Nabunken data.

Each cell gives observed width / supplied growth-curve value, both in mm. Index = observed width divided by curve value. Curve fitting itself is not part of this exercise.

Year	Tree A	Tree B	Tree C
1998	2.26560 / 1.88800	1.92698 / 1.60582	1.09175 / 1.21306
1999	1.65469 / 1.83854	1.24045 / 1.55056	1.18460 / 1.18460
2000	1.96986 / 1.79078	1.94705 / 1.49773	—

Calculate each index to two decimal places. Average only available trees with equal weights, and record n. What happens if missing C in 2000 is replaced by zero?

Check

Year	Indices A, B, C	Mean	n
1998	1.20, 1.20, 0.90	1.10	3
1999	0.90, 0.80, 1.00	0.90	3
2000	1.10, 1.30, missing	1.20	2

Filling the missing value with zero gives a mean of 0.80, inventing a growth decline that was not observed. Do not count multiple radii from one tree as independent trees.

Return to the measured data

In the original 29ho018 CSV, 32 means 0.32 mm. The first seven widths sum to 2.79 mm. This is a sum of published measurements, not a new measurement from the printed photo. Check the original scale and measurement path; document positions and uncertainty if remeasuring.

29ho018: source image and measurements